

Flat Rolling Slab Model PyRoLL Plugin

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The PyRoLL plugin `pyroll-flat-rolling-slab-model` calculates the roll-force and roll-torque as well as the mean neutral plane location for a roll-pass. Since von-Karman originally developed the strip or slap theory for flat rolling, the grooved pass is not considered directly. The solution is therefore derived for an equivalent flat pass, which makes the plugin's accuracy heavily depended on the equivalent flat pass plugins.

The plugin provides three solvers, covering the full range from thick slabs to metal foils, and selects between them automatically for each roll pass. All three are elastic-plastic (Hooke's-law entry and exit zones bound the plastic zone, rather than assuming the whole contact is plastic from first contact) and use Bay and Wanheim's smoothed mixed Coulomb/sticking friction law throughout - there is no rigid-plastic or pure-Coulomb model in this plugin:

- a mixed-friction solver for medium passes (section 2), that keeps von-Kármán's equilibrium equation and closes it with the plain Mises relation;
- an elastic-plastic layer model (section 3), used for thick passes, that adds through-thickness resolution into multiple layers and full thermal coupling on top of the same construction;
- a full elastic-contour foil-rolling model (section 4), used once elastic roll flattening becomes so severe that a circular roll-gap assumption is no longer valid.

section 5 explains the criteria used to select between the three, and section 8 lists the hooks the plugin defines. A fourth model, Orowan's own 1943 hot-rolling theory (section 6), is also implemented but deliberately kept outside this automatic selection - section 6 explains why and how it is used instead.

1 Von-Kármán's equilibrium equation

The current model is based on the classic elementary theory of plasticity, which is a well known approach of modelling the stress distributions in flat rolling. The elementary theory of plasticity for flat rolling is also known as strip or slap theory, because the roll gap is divided in rolling direction in infinitesimal narrow strips, as shown in Figure 1, on

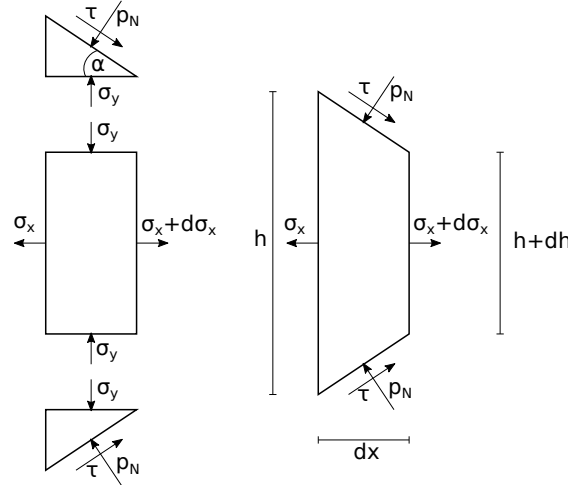


Figure 1: Stripe element of elementary theory of plasticity

which the force balance is built. In thickness direction the strip's height is equal the local height of the roll gap. The deformation of each strip is always plane and parallelepipedic. The strip's material properties are constant in thickness direction and may vary in rolling direction. Therefore, the vertical stress is constant in thickness direction and no shear deformation can be considered.

The roll surface is described as a circular arc, since elastic roll deformation is neglected in this model (see section 3 and section 4 for the elastic-plastic and elastic-contour models, which lift this restriction). So the roll gap height can be described by Equation 1. The rolls are here considered to be rigid. Also, the rolls are considered to be symmetric, meaning of equal radius R_W and rotational frequency n_W .

$$h(x) = s + 2 * \left(R_W - \sqrt{R_W^2 - x^2} \right) \quad (1)$$

The model needs additional approaches for the yield stress, and the friction between rolls and workpiece. The force balance at the stripe element leads to the well known Kármán differential equation [1, 2] as shown in Equation 3, which is an ordinary differential equation for the horizontal stress σ_x in x and can be solved easily by numerical methods. A suitable yield criterion gives the connection between σ_x and σ_y ; every model in this plugin (section 2–section 6) integrates Equation 3 unchanged, differing only in the friction law closing τ_R (below), the algebraic relation closing σ_x and σ_y , and whether the strip is treated as one homogeneous cross-section or several through-thickness layers.

$$0 = \sigma_x \frac{dh}{dx} + h \frac{d\sigma_x}{dx} + 2\tau_R - 2p_N \tan(\alpha) \quad (2)$$

$$p_N = -\sigma_y - \tau_R \tan(\alpha) \quad (3)$$

Every model in this plugin bounds the plastic zone integrating Equation 3 with Hooke's-law elastic entry and exit zones (rather than assuming the whole contact is plastic from

the first point of contact), and closes the friction shear stress τ_R with Bay and Wanheim's mixed Coulomb/sticking law (Equation 5) rather than plain Coulomb friction - section 2 and section 6 share this construction in `_elastic_plastic_common.py`, differing only in the closure relating σ_x and σ_y .

Equation 3 is itself only the *static* force balance (Plastostatik) at the strip element - it neglects the inertia of the material accelerating through the roll gap and any direct dependence of flow stress on strain rate beyond what is passed into `flow_stress_function`. Troost [3] extends the same elementary strip-element theory (citing von Kármán, Siebel and Orowan as its origin) to a "plastokinetic" one: adding the volume element's inertial force (via d'Alembert's principle) and the strain-rate/temperature dependence of flow stress lets Gl. (5)-(11) of that paper be recast as a *dynamic* flow stress k_f^* that folds both effects into an otherwise unchanged static solution - existing solutions of Equation 3 (and its approximations) carry over unmodified once k_f is replaced by k_f^* . Troost's own worked example found the resulting correction small in practice ($k_f^* - k_f$ on the order of a few MPa, and roll force varying by only about $\pm 10\%$ across a 1 m/s-to-50 m/s exit-velocity range for a first, unworked pass) and dominated in his results by the velocity-dependence of the friction coefficient rather than by inertia itself.

The inertial half of Troost's correction (`dynamic_flow_stress.py`) is implemented as an optional add-on, gated by the `dynamic_flow_stress_correction_enabled` hook (default `False`) and applied identically at every flow-stress evaluation in all four solvers - the height ratio $h_0/h(x)$ his eq. (12) needs is recovered from the equivalent strain already computed at each call site ($\varphi_v = 2/\sqrt{3} \cdot \ln(h_0/h(x))$), so no extra geometry plumbing is required. The reference velocity feeding the "Staudruck" $q_{st,0} = \frac{1}{2}\rho U_0^2$ is the roll's own peripheral speed $2\pi nR$, rather than a self-consistently slip-derived strip velocity - Troost's own worked example likewise treats speed as an independent input rather than solving for it jointly with the pressure distribution, and the correction's magnitude does not warrant that added rigor. Left off by default because, as the numbers above show, it stays under 1% of a typical cold-rolling flow stress below about 10 m/s and only reaches a low single-digit percentage at speeds above roughly 75 m/s, well beyond this plugin's validated range; it does not implement the strain-rate/temperature half of Troost's k_f^* (already covered by whatever dependence `flow_stress_function` itself has on `strain_rate`) or his refined friction-velocity dependence. Like Ekelund's redundant-shear correction (section 3), the rest of Troost's plastokinetic theory beyond this inertial term is noted here as a documented, deliberately out-of-scope refinement to the same underlying slab theory.

2 Mixed-friction model for medium passes

Pure Coulomb friction, integrated all the way across the contact as in Equation 3, demands ever-increasing shear traction toward the roll gap centre; once a pass is thick or high-friction enough, this exceeds what the material can actually transmit and produces an unrealistic, unbounded pressure spike. `karman_mixed_friction_solver.py` implements `KarmanMixedFrictionSolver`, which keeps von-Kármán's force balance exactly as in Equa-

tion 3, closes σ_x and σ_y with the plain Mises relation $\sigma_x = \sigma_y + 2k_f/\sqrt{3}$, and caps the friction shear stress at the material's sticking (shear-yield) limit once Coulomb friction would exceed it, giving a friction hill that saturates ("sticking") rather than diverging, using the same smoothed Bay and Wanheim [4] mixed Coulomb/sticking law as the layer model (section 3, Equation 5), applied to the whole (single, homogeneous) cross-section rather than per layer. Hooke's-law elastic entry and exit zones bound the plastic zone on both sides (`_elastic_plastic_common.py`, shared with section 6); unlike the layer model, it has no thermal coupling and treats the cross-section as a single homogeneous layer - it is kept as its own class rather than dispatching to the layer model with a single layer, so that a pass not needing thermal coupling or per-layer resolution does not pay for either.

This is *not* Orowan's own 1943 slab theory [5], despite addressing the same underlying observation (that pure Coulomb friction becomes unrealistic once passes are thick or high-friction enough). Orowan's actual model replaces the straight strip-element boundaries of Figure 1 with circular arcs, centred on the roll-gap symmetry line and meeting the roll surfaces at right angles, and corrects the horizontal force balance with an inhomogeneity function $\omega(a, \alpha)$ - an integral over the arc's polar angle - that captures genuinely non-uniform through-thickness deformation directly, rather than refining the friction law on top of the same homogeneous-strip geometry as section 1. Orowan himself noted the general theory was too complex for practical calculation at the time; later simplifications (e.g. Sims' application of it to hot rolling with sticking friction throughout) are reported to give unrealistic results (forming efficiencies above 100%) under some conditions. section 6 implements Orowan's own inhomogeneity-function theory as a separate, explicitly-selected model, rather than as a further refinement dispatched automatically alongside the models above.

3 Elastic-plastic layer model for thick passes

For thick passes, the models above miss two further effects that become significant as the relative pass thickness grows: an elastic entry/compression zone and elastic recovery zone at exit (rather than the whole contact being plastic immediately), and, once the strip is thick enough relative to the contact length, meaningfully different flow stress between the surface (which alone feels roll-contact friction) and the core through the pass. `layer_solver.py` implements `LayerRollingSolver`, a layer-count-parametrized model following Weiner [6] (based on his *RollingHomogeneousElasticPlastic* and *RollingLayerModelElasticPlastic* implementations), which addresses both:

- the strip is discretized into N through-thickness layers (used here for $N > 1$, the "thick slab" regime; $N = 1$ collapses to a single homogeneous slab equivalent to section 2's own construction, but `KarmanMixedFrictionSolver` is kept as its own class rather than dispatching to the layer model with $N = 1$, since it needs neither thermal coupling nor per-layer resolution);
- each layer starts elastic and yields once its own local Mises criterion is met, giving genuine elastic entry and exit zones per layer instead of a single sharp elastic/plastic

split;

- friction at the roll interface, and between adjacent layers, follows the same mixed Coulomb/sticking law as section 2 after Bay and Wanheim [4] (Equation 5), transitioning smoothly from Coulomb sliding to a sticking-limited shear stress as normal pressure grows;
- deformation heat, friction heat, conduction between neighbouring layers (and to the rolls), and convective transport are solved simultaneously with the stress equations, so each layer's temperature - and hence its flow stress - evolves individually through the pass.

$$p_{N,\text{Coulomb}} = \frac{-\sigma_y}{1 + \mu \tan(\alpha) \text{sign}}, \quad p_{N,\text{crit}} = \frac{m k_f}{\mu \sqrt{3}} \quad (4)$$

$$\tau = \left[\mu p_{N,\text{Coulomb}} (1 - \Phi) + \frac{m k_f}{\sqrt{3}} \Phi \right] \text{sign}, \quad \Phi = \frac{1}{\pi} \arctan \left(\frac{p_{N,\text{Coulomb}} - p_{N,\text{crit}}}{c p_{N,\text{crit}}} \right) + \frac{1}{2} \quad (5)$$

where m is the sticking friction coefficient (`friction_stiction_coefficient`), `sign` the (regularized) direction of relative sliding, and c a smoothing parameter (`pressure_smoothness`) controlling how sharply the transition Φ switches from Coulomb ($\Phi = 0$) to sticking ($\Phi = 1$) around the critical pressure $p_{N,\text{crit}}$.

Roll flattening is deliberately out of scope here, as it is for the other solvers in this plugin: all three simply read `Roll.working_radius` once rather than deriving a flattened radius themselves. A separate plugin providing a Hitchcock [7] (or other) flattened radius via that hook is therefore picked up transparently by every solver in this plugin, with no special-casing needed; this plugin's own dispatch criterion, the contact-length-to-mean-thickness ratio implemented in `condition.py` (`contact_length_over_mean_thickness`), exists solely to *classify* a pass for dispatch (section 5), and is never fed back into a calculation.

Because each pass only sees the incoming profile's own scalar strain and temperature (`pyroll-core` has no native concept of a through-thickness state carried from one pass to the next), all layers start uniform; the elastic entry zone therefore has no mechanism to break symmetry between layers (matching the reference implementation, which defines but never uses a redundant-shear correction after Ekelund [8]). Layers still diverge from the first plastification onward, since only the outermost layers feel roll-contact friction directly while inner layers feel the weaker inner-friction law, and the resulting temperature and flow-stress differences compound through the rest of the pass.

4 Full elastic-contour foil-rolling model

Once elastic roll flattening becomes severe enough, even a scalar (Hitchcock) roll-flattening correction to `Roll.working_radius` is no longer accurate, because the flattened roll-gap

shape stops being well approximated by a single effective radius. `foil_solver.py` implements `FoilRollingSolver`, following Mauk and Overhagen [9], which instead computes the true, generally non-circular, deformed roll-gap shape directly: the normal-pressure distribution along the contact is superposed with elastic half-space (Boussinesq/Johnson [10]) influence coefficients to obtain the roll surface’s own elastic deflection, iterated to convergence with the strip’s own elastic-plastic stress solution (elastic entry, plastic slip, an optional sticking zone, plastic slip, elastic exit).

This converged shape - against the rigid, circular baseline it started from - is exposed as `roll_contour`, a dataframe indexed by roll-gap position x with columns `gap_height` (the flattened shape) and `rigid_gap_height` (the undeformed circular arc it is compared against); `report.py`’s `foil_rolling_contour_plot` renders it automatically whenever `FoilRollingSolver` is the dispatched solver. Figure 2 shows it for the 0.1mm-to-0.077mm steel-foil pass of Table 1 (Table 5 of the report): the two curves are visually close - a reminder that the flattening is a small correction to an already-small gap, not a gross reshaping - but the deviation panel below makes the effect visible on its own scale: a few micrometres, asymmetric about the roll-gap centre (the entry side flattens more than the exit side here), and consistently load-bearing rather than numerical noise, since it is exactly this deviation that the influence-coefficient/pressure iteration was solving for to begin with.

The deviation’s magnitude is governed by roll stiffness, not just gauge: Figure 3 repeats the same pass with a ceramic work roll (StarCeram N8000, $E = 310$ GPa) in place of steel ($E = 210$ GPa) - the report’s own headline comparison (Table 1, Abb. 9). The stiffer ceramic roll flattens noticeably less at its peak and needs a correspondingly lower roll force, exactly the effect `test_foil_solver_converges_and_reduces_force_with_stiffer_roll` checks for.

Reproducing the report’s own Abb.9 numbers literally ($k_f = 1000$ N/mm², 25mm rolls, 50% reduction, zero tension - a considerably more aggressive point than the pass above) exposed a genuine numerical limitation of the outer elastic-flattening iteration rather than a wrong result. At that reduction the converged roll-gap shape develops a near-degenerate contour (a shallow local kink from an incipient sticking sub-zone right beside the true global minimum), and chasing why the neutral-point search failed there improved the solver’s robustness in three independently validated ways: `_gap_minimum` now locates the roll-gap’s global minimum by height value rather than by the first derivative sign change, so a shallow kink preceding the true minimum can no longer collapse the neutral-point search domain onto a spurious, too-early bound; the outer loop’s relaxation factor is now adaptively halved whenever a step fails to shrink the iterate-to-iterate change, breaking limit cycles a fixed damping factor cannot; and the plain relaxed update is now Anderson(m) mixing [11], extrapolating from the several most recent iterates’ residuals via a small least-squares fit rather than blending only the latest one - the standard fix for a Picard iteration whose convergence is dominated by a single near-degenerate direction, exactly what a small but load-bearing elastic-flattening correction against a much larger rigid baseline produces. Each accelerated step is re-validated through the full entry/neutral-point/zone-chain physics before being accepted, falling back to a shrunk plain-relaxed step (and ultimately raising, rather than silently continuing, if even that

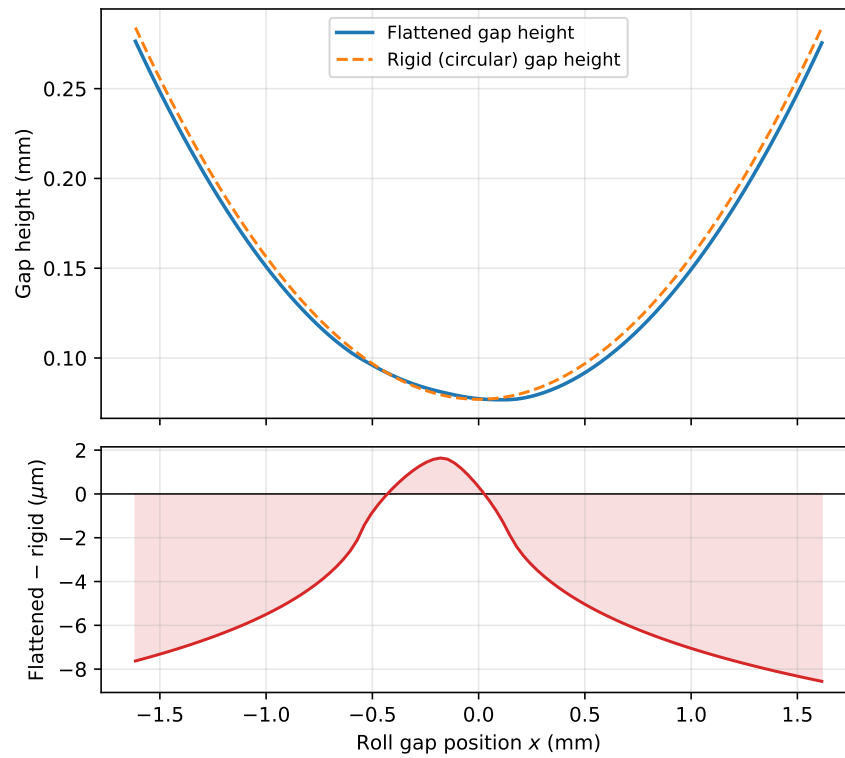


Figure 2: Flattened vs. rigid (circular) roll-gap contour for a representative foil pass, with the deviation between them shown on its own scale below.

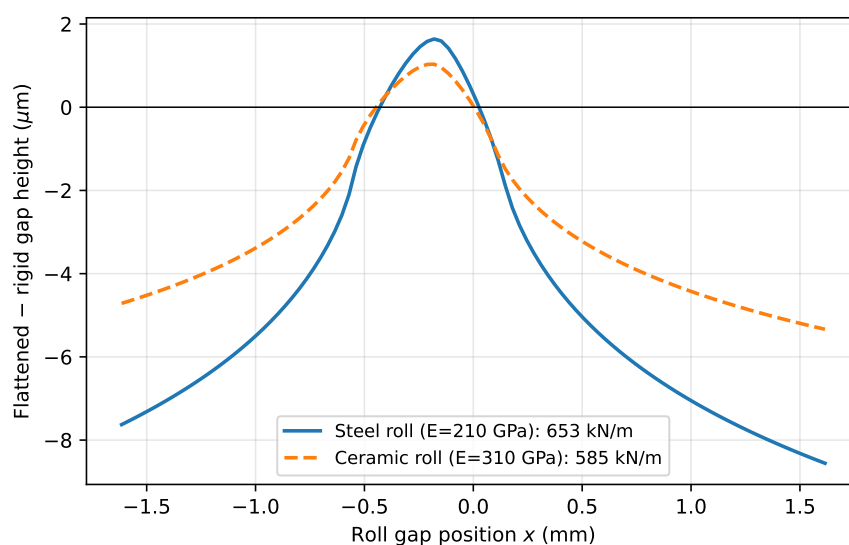


Figure 3: Roll-gap deviation for the same pass as Figure 2, comparing a steel and a ceramic work roll - the stiffer roll flattens less and needs a lower roll force for the same reduction.

fails), since an unvalidated extrapolated shape can otherwise land somewhere the neutral-point search cannot bracket at all. That per-step validation is not free: it re-runs the same physics used to build the plain update, so the full `tests/test_foil_solve.py` slow suite’s wall-clock cost grew from about 50 minutes to a little over two hours once this was added - an accepted cost for a solver that no longer stalls or crashes on harder passes.

Even with all three fixes, Abb. 9’s literal parameters still fail: even an infinitesimal step away from an already-accepted shape breaks the neutral-point bracket there, indicating a genuine structural fragility of the zone-chain search at this particular, unusually aggressive reduction/tension combination, rather than merely slow convergence. Reproducing Abb. 9’s own 690 kN (steel) vs. 521 kN (ceramic) force values exactly is accordingly left as a known limitation rather than pursued further; Figure 3 and `test_foil_solver_converges_and_reduces_force` already reproduce the report’s qualitative headline result - a stiffer roll flattens less and needs a lower force for the same reduction - on a less extreme, convergent pass, and that is what this plugin is validated against.

Gauge itself also matters, independent of roll stiffness: Figure 4 holds the roll and the relative schedule fixed (23% reduction, $\mu = 0.1$, the same tension-to-flow-stress ratios) and only varies the incoming thickness, plotting the deviation *relative to the local rigid gap height* (since the absolute gap, and hence the absolute deviation, shrinks with gauge too) - a thinner incoming gauge (0.05 mm vs. 0.10 mm) shows a substantially larger relative deviation (up to about 11% vs. 5% here), confirming `test_thinner_foil_flattens_more_than_thicker_foil`’s finding: the same absolute roll stiffness and force make up a bigger fraction of an already-smaller gap as the foil gets thinner.

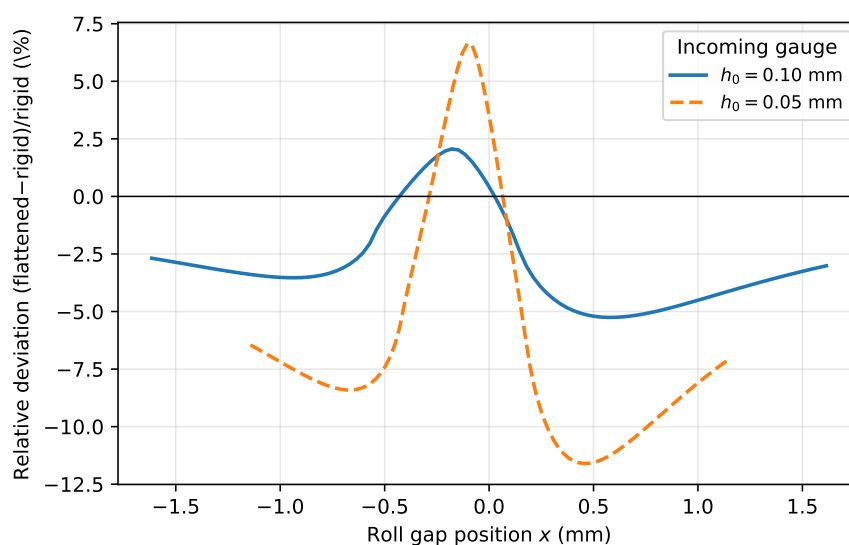


Figure 4: Relative roll-gap deviation $((\text{flattened} - \text{rigid})/\text{rigid})$ for two incoming gauges at the same relative reduction, friction and tension ratios - the thinner foil flattens proportionally more.

5 Selecting between the three models

`roll_pass.py` wires the three solvers together behind a single hook, `karman_solution`, so that `roll_force`, `Roll.roll_torque`, `Roll.neutral_point` and the profile velocity hooks work unchanged regardless of which solver actually ran:

A single quantity drives the whole decision: the classical contact-length-to-mean-thickness ratio L_d/H_m (`contact_length_over_mean_thickness` in `condition.py`), a cheap, purely geometric calculation needing no solver to be run first.

1. if L_d/H_m is below `layer_model_ld_hm_limit` (default 1.0), through-thickness deformation is inhomogeneous enough that the layer model of section 3 is used;
2. if L_d/H_m is above `foil_rolling_ld_hm_limit` (default 10.0), the pass is thin enough relative to the contact length that elastic roll flattening can no longer be neglected, and the foil-rolling model of section 4 is used;
3. otherwise, the mixed-friction model (section 2) is used for medium passes.

Earlier versions of this plugin used the Hitchcock flattened-to-nominal radius ratio r'/r (Mauk and Overhagen [9] give $r'/r \geq 2$ as the point where a circular roll-gap assumption is no longer trustworthy) for the foil decision instead, requiring an extra `KarmanMixedFrictionSolver` solve purely to obtain a force estimate for it; using L_d/H_m for all three decisions removes that extra solve and keeps the whole dispatch to one consistent, purely geometric criterion.

Both limits are themselves hooks (Table 4) and can be overridden per pass, including forcing a particular model unconditionally. Since all three models are elastic-plastic, the roll and profile elastic properties (`elastic_modulus`, `poissons_ratio`) must be set for every pass - there is no rigid-plastic fallback for passes that omit them; a pass without them simply fails to solve.

Table 1 summarizes all four models by their underlying stress-strain treatment, friction law, and the thickness regime each is intended for - the three dispatched automatically by the criteria above, plus `OrowanSolver` (section 6), which is not.

6 Orowan's (1943) hot-rolling model

Every model in section 1–section 4 keeps von-Kármán's original strip geometry (Figure 1): straight, vertical strip boundaries, a homogeneous stress state through the strip's thickness at each x , and (in section 3) at best a coarse, discretized approximation of through-thickness variation via a handful of layers. Orowan's 1943 theory [5] - as reconstructed here from Overhagen's account of it [12, sec. 4.4.1.2], the original 1943 paper not being available for direct comparison - instead changes the strip element's own geometry: circular arcs, centred on the roll-gap symmetry line and meeting the roll surfaces at right angles, replace the straight vertical boundaries, with a polar coordinate system on each arc carrying an inner tangential stress $\sigma_{t,i}$, radial stress $\sigma_{r,i}$ and shear stress τ_i . This directly represents the non-uniform, redundant through-thickness deformation Orowan observed rolling plasticine specimens, rather than approximating it through a handful of discrete layers as section 3 does.

`orowan_solver.py` implements `OrowanSolver`: the horizontal force balance is von-Kármán's own equation, unchanged from Equation 3's first line, but the relation between σ_x and σ_y that closes it is no longer the plain Mises offset - it is Orowan's own closure, correcting for the inhomogeneous deformation the circular-arc geometry captures:

$$\sigma_x = \sigma_y + \frac{2k_f}{\sqrt{3}} \left[\omega_O(\alpha) \mp \frac{1}{2} \left(\frac{1}{\alpha} - \frac{1}{\tan \alpha} \right) \right] \quad (6)$$

$$\omega_O(\alpha) = \frac{1}{\sin \alpha} \int_0^\alpha \sqrt{1 - \left(\frac{\theta}{\alpha} \right)^2} \cos \theta \, d\theta = \frac{\pi J_1(\alpha)}{2 \sin \alpha} \quad (7)$$

where θ is the polar angle within an arc and J_1 the order-1 Bessel function of the first kind. The "-" sign in Equation 6 applies on the entry (Vorlauf) side, "+" on the exit (Nachlauf) side, mirroring the sign convention already used for τ_R in Equation 3. The inhomogeneity function ω_O ranges from 1 (frictionless limit) down to $\pi/4 \approx 0.785$ as $\alpha \rightarrow 0$ (full sticking) and depends only weakly on α otherwise [12, Abb. 4.4/4] - it is what makes Equation 6 a genuine correction to, rather than a mere re-derivation of, the plain Mises relation $\sigma_x = \sigma_y + 2k_f/\sqrt{3}$ every other model in this plugin uses. Orowan's general theory pairs Equation 6 with a sliding-friction counterpart (his Gl. 4.4/5) and a friction-ratio parameter $a_O = 2\mu\sigma_N/k_f \in [0, 1]$ blending between the two; rather than rederiving Gl. 4.4/5, `OrowanSolver` instead keeps Equation 6's sticking-limit structure

Table 1: Overview of stress-strain and friction models, with recommended thickness regime.

Model	Stress-strain model	Friction model	Recommended regime
Karman-Mixed-Friction-Solver	elastic-plastic, homogeneous cross-section	Bay/Wanheim mixed Coulomb/-sticking	medium gauge: $1 \leq L_d/H_m \leq 10$ (defaults) - the plugin's default once neither condition below triggers
Layer-Rolling-Solver	elastic-plastic, N through-thickness layers, full thermal coupling	Bay/Wanheim mixed (roll contact & inter-layer)	thick slabs: $L_d/H_m < 1$ (default), i.e. contact length small relative to mean thickness
Foil-Rolling-Solver	elastic-plastic, homogeneous cross-section, elastic (non-circular) roll-gap contour	Coulomb with optional sticking zone	foil rolling: $L_d/H_m > 10$ (default), i.e. contact length large relative to mean thickness - our own 0.1mm-to-0.077mm steel-foil case (Figure 2) reaches $L_d/H_m \approx 6$, and thinner/higher-friction variants comfortably exceed 10; industrial foils are typically well under 0.1 mm and correspondingly further into this regime
Orowan-Solver	elastic-plastic, single homogeneous cross-section, circular-arc inhomogeneity correction	Bay/Wanheim mixed	not gated by L_d/H_m - always selected explicitly (section 8). A hot-rolling alternative to KarmanMixedFriction-Solver/LayerRollingSolver , typically overlapping the medium-to-thick gauge range

($a_O \equiv 1$) throughout and layers Bay and Wanheim's mixed Coulomb/sticking law [4] (Equation 5, the same law section 2 and section 3 use) on top of it - see below for exactly how the two combine. Elastic entry and exit zones bound the plastic zone on both sides (`_elastic_plastic_common.py`, shared with section 2), so that (unlike Orowan's own rigid-plastic 1943 paper) the whole contact is not assumed plastic from first contact.

Reconstructing Equation 6–Equation 7 from [12] required three specific interpretive choices, each checked as far as the available material allows but not verifiable against Orowan's original 1943 paper directly - documented in full in `orowan_solver.py`'s module docstring, and summarized here:

- Overhagen's own Gl. 4.4/7 prints ω_O 's integration limit as a_O rather than α ; Equation 7 integrates to α instead, since only that choice reproduces the two values his own Abb. 4.4/4 plots ($\omega_O \rightarrow 1$ as $a_O \rightarrow 0$; $\omega_O \rightarrow \pi/4$ as $a_O \rightarrow 1, \alpha \rightarrow 0$ - both checked against Equation 7's closed form in `tests/test_orowan_solve.py`), and the a_O -limit version is not even well-defined in general (its square root goes negative once $a_O/\alpha > 1$, the common case at realistic rolling angles and high a_O);
- k_f in Equation 6 is $2\times$ the uniaxial flow stress `flow_stress_function` returns elsewhere in this plugin (Overhagen's own source equations use a bare $k_f/\sqrt{3}$ offset, consistent with an older, Tresca-flavoured plane-strain convention for k_f predating this plugin's Mises one) - chosen specifically so Equation 6 reduces to the ordinary Mises relation in the frictionless limit, keeping the elastic-zone yield-onset criterion (still the plain Mises condition) and the plastic-zone closure continuous with each other;
- the entry/exit sign of the $(\frac{1}{\alpha} - \frac{1}{\tan \alpha})$ term is a best-effort assignment: Overhagen's source prints it as " $\mp$ " without enough surrounding context in the excerpt available to fix which sign belongs to which side independently.

None of these affect ω_O 's qualitative role or Equation 6's overall structure; they are exactly the kind of detail worth re-checking against Orowan's 1943 paper directly if it becomes available.

The two closed-form checks above (`tests/test_orowan_solve.py`) only confirm the corrected upper integration limit at the two corners of Overhagen's own Abb. 4.4/4 ($a_O \rightarrow 0$ and $a_O \rightarrow 1, \alpha \rightarrow 0$). A stronger check is to evaluate $\omega_O(\alpha, a_O)$ from Equation 7 at the general $a_O \in [0, 1]$ Overhagen's Gl. 4.4/7 is actually stated for - substituting $t = \theta/\alpha$ gives the equivalent, quadrature-friendly form

$$\omega_O(\alpha, a_O) = \frac{\alpha}{\sin \alpha} \int_0^1 \sqrt{1 - (a_O t)^2} \cos(\alpha t) dt, \quad (8)$$

implemented as `omega_orowan_general` in `orowan_solver.py` purely for this validation - `OrowanSolver` itself still only ever needs the $a_O = 1$ slice via the mixed-friction blend described below. Equation 8 reduces to Equation 7's closed form at $a_O = 1$ (checked to match `omega_orowan_sticking` to machine precision) and to $\omega_O(0, a_O) = \frac{1}{2} \left(\sqrt{1 - a_O^2} + \arcsin(a_O)/a_O \right)$

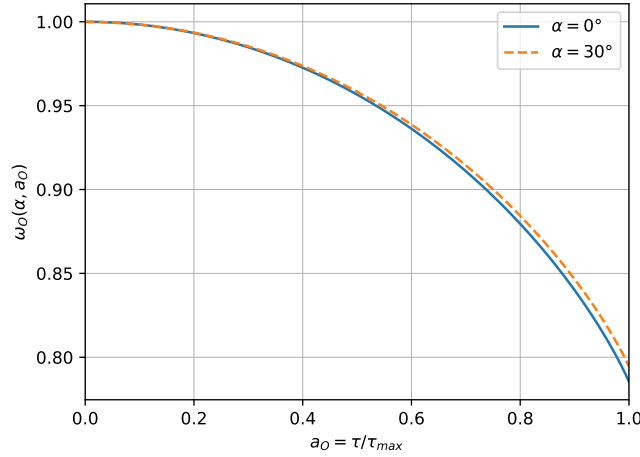


Figure 5: Reproduction of Overhagen's Abb. 4.4/4 [12]: the inhomogeneity function $\omega_O(\alpha, a_O)$ from Equation 8, evaluated across the full friction-ratio range $a_O \in [0, 1]$ at $\alpha = 0^\circ$ and $\alpha = 30^\circ$.

at $\alpha = 0$ by l'Hôpital's rule on $\cos(\alpha t) \rightarrow 1$ - both cross-checked in `tests/test_orowan_solve.py`. Figure 5 reproduces Abb. 4.4/4 in full across the whole a_O range at both angles it plots, rather than just its two derivable corners, and matches its shape and reported values closely: ω_O falls monotonically from 1 at $a_O = 0$ to about 0.785-0.79 at $a_O = 1$, with the $\alpha = 30^\circ$ curve consistently, but only slightly, above the $\alpha = 0^\circ$ one - exactly the "depends only weakly on α " behavior Overhagen's own text states.

Generalizing Equation 6 to mixed friction, rather than Orowan's own a_O -dependent sliding branch (his Gl. 4.4/5), keeps its structure unchanged and generalizes only ω_O : the sticking value $\omega_O(\alpha)$ of Equation 7 is linearly blended toward the frictionless limit 1 using Φ , the same smooth Coulomb-to-sticking transition variable Equation 5 already computes for the friction law itself,

$$\omega_{O,\text{eff}}(\alpha, \Phi) = (1 - \Phi) \cdot 1 + \Phi \cdot \omega_O(\alpha) \quad (9)$$

which is a deliberate, pragmatic hybrid rather than a literal implementation of Gl. 4.4/5 - justified by ω_O 's own weak dependence on its arguments (noted above), so linearly interpolating between its two known closed-form endpoints, weighted by the regime indicator already driving the friction law, avoids needing the general, quadrature-only $\omega_O(\alpha, a_O)$ for $a_O \notin \{0, 1\}$. Since Φ itself depends on σ_y - the quantity Equation 6 solves for - `OrowanSolver` iterates Equation 6 and Equation 9 to a fixed point (converging in a handful of iterations, given how weakly ω_O varies). Bay and Wanheim's law is well-behaved as pressure $\rightarrow 0$ (unlike assuming sticking friction where contact pressure is still essentially zero), so it applies unmodified in the elastic entry/exit zones too, exactly as section 2 and section 3 already do.

`OrowanSolver` is not wired into the `karman_solution` dispatch hook of section 5: it must be instantiated directly, `OrowanSolver(roll_pass=roll_pass)`, for a pass whose profile has

Table 2: Orowan’s model compared to the plugin’s other solvers.

Model	Strip geometry	Friction law	Elastic zones	Thermal
KarmanMixedFrictionSolver	straight, homogeneous	Bay/Wanheim mixed	entry exit	& none
LayerRollingSolver	straight, N discrete layers	Bay/Wanheim mixed	entry exit, layer	& full per
FoilRollingSolver	straight, homogeneous, elastic roll contour	Coulomb + optional sticking	entry exit	& none
OrowanSolver	circular arc, continuous inhomogeneity	Bay/Wanheim mixed	entry exit	& none

already been solved. This is a deliberate scope decision, not an oversight - the model is a genuinely different, orthogonal refinement (curved-strip inhomogeneity) from the axis section 5 already selects along (elastic flattening, then contact-length-to-thickness), it applies only to a single homogeneous cross-section (no layer-count parametrization, unlike section 3), and it has no thermal coupling (flow stress is evaluated at the incoming profile’s own scalar temperature throughout, like section 1 and section 2). Table 2 summarizes how it relates to the other models in this plugin.

7 Comparison of the models

Because **KarmanMixedFrictionSolver**, **LayerRollingSolver** and **OrowanSolver** all integrate the same equilibrium equation (Equation 3) with the same mixed friction law (Equation 5), differing only in their σ_x/σ_y closure and (for the layer model) through-thickness resolution, they can be run side by side on the same pass and compared directly - unlike **FoilRollingSolver**, whose elastic roll-gap contour only differs meaningfully from a rigid one once $L_d/H_m > 10$ (section 5), well outside the gauge range the other three are meant for.

Table 3 and Figure 6 compare all four on the same representative hot-rolling pass ($h_0 = 15$ mm, gap 12 mm, work-roll radius 160 mm, $\mu = 0.35$, constant $k_f = 100$ MPa, no tension) - the same pass used in this plugin’s own test suite. All four agree on roll force to within about 10% and on the friction-hill shape (Figure 6): **OrowanSolver** predicts a somewhat lower peak pressure and force than **KarmanMixedFrictionSolver**, since its inhomogeneity function ω_O (Equation 7) is ≤ 1 and so only ever reduces the Mises offset relative to the plain closure; **LayerRollingSolver** with a single layer ($N = 1$) falls in between, close to but not identical to **KarmanMixedFrictionSolver**, since it is a materially different implementation (thermally coupled, its own elastic-plastic constitutive

Table 3: Solver outputs for the same representative hot-rolling pass.

Model	Force (kN/m)	Torque (kN)	x_n (mm)	Peak p_N (MPa)
KarmanMixedFrictionSolver	3365	34.7	-8.15	187.9
OrowanSolver	3038	31.5	-8.25	165.6
LayerRollingSolver ($N = 1$)	3237	33.6	-8.44	168.6
LayerRollingSolver ($N = 5$)	3194	33.3	-10.46	163.4

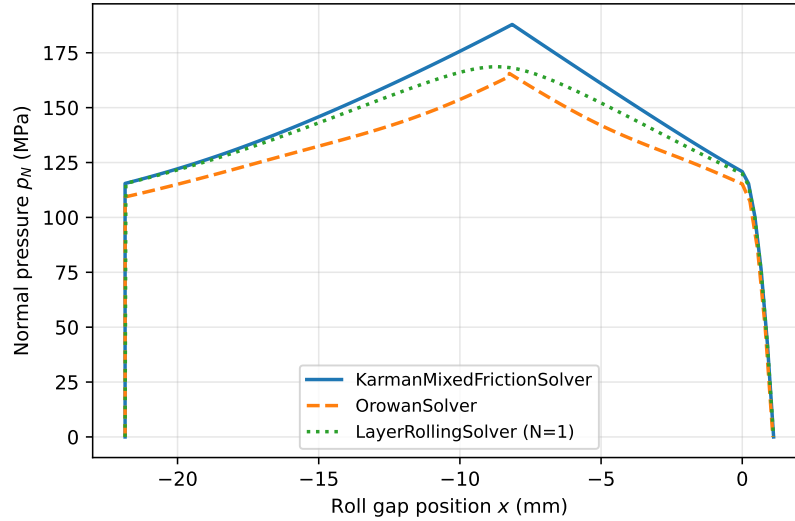


Figure 6: Normal-pressure distribution across the roll gap for the same pass as Table 3, comparing `KarmanMixedFrictionSolver`, `OrowanSolver` and `LayerRollingSolver` ($N = 1$).

formulation) that happens to collapse to the same physical limit rather than sharing code with it (section 2); and `LayerRollingSolver` with five layers ($N = 5$) sits close to its own $N = 1$ result, as expected since this particular pass is not actually in the thick-slab regime (L_d/H_m comfortably above the default limit of 1) - the layer count only matters once through-thickness deformation is genuinely inhomogeneous.

8 Usage instructions

The plugin can be loaded under the name `pyroll_flat_rolling_slab_model`.

An implementation of the `roll_force` hook on `RollPass` is provided. Furthermore, an implementation of the `roll_torque` and `neutral_point` hooks on `RollPass.Roll` is provided.

Additionally, hooks on `RollPass` are defined, which are used in the calculation, as listed in Table 4. The hooks `back_tension`, `front_tension` and `coulomb_friction_coefficient`

Table 4: Hooks specified by this plugin.

Hook name	Meaning
<code>coulomb_friction_coefficient</code>	Coulomb's friction coefficient μ
<code>back_tension</code>	Back tension of the roll pass $\sigma_{x,0}$
<code>front_tension</code>	Front tension of the roll pass $\sigma_{x,1}$
<code>karman_solution</code>	Solver object (of one of the three models) with solution values as attributes
<code>foil_rolling_condition</code>	Whether the foil-rolling model (section 4) is used
<code>foil_rolling_ld_hm_limit</code>	L_d/H_m ratio above which the foil model is used (default 10.0)
<code>thick_slab_condition</code>	Whether the multi-layer, thick-slab configuration (section 3) is used instead of <code>KarmanMixedFrictionSolver</code> (section 2)
<code>layer_model_ld_hm_limit</code>	L_d/H_m ratio below which the multi-layer configuration is used (default 1.0)
<code>layer_model_layer_count</code>	Number of through-thickness layers on the thick-slab path (default 5)
<code>friction_stiction_coefficient</code>	Sticking friction coefficient m feeding the mixed friction law (Equation 5) of <code>KarmanMixedFrictionSolver</code> and <code>LayerRollingSolver</code> (default 0.8)
<code>roll_heat_transfer_coefficient</code>	Heat transfer coefficient at the roll/material interface (default 6000 W/(m ² K))
<code>dynamic_flow_stress_correction_enabled</code>	Whether Troost's inertial flow-stress correction (section 1) is applied (default <code>False</code>)

have to be set and adjusted individually; since all three dispatched models are elastic-plastic, `elastic_modulus` and `poissons_ratio` must also be set on both `RollPass.Roll` and `RollPass.InProfile` for every pass (see section 5) - there is no fallback for passes that omit them - and the layer model's thermal coupling additionally needs `specific_heat_capacity`, `thermal_conductivity` and `density` set on both.

`OrowanSolver` (section 6) is not reachable through `karman_solution`, since it is not part of the automatic dispatch of section 5; call it directly once the pass has been solved, e.g. `OrowanSolver(roll_pass=roll_pass)` after `PassSequence([roll_pass]).solve(in_profile)`, then read `roll_force_per_unit_width`, `roll_torque_per_unit_width`, `neutral_plane_position` and `solution` off the returned instance directly, the same attributes the other three solvers expose.

`FoilRollingSolver` additionally exposes `roll_contour` (section 4, Figure 2), the converged flattened roll-gap shape against its rigid circular baseline - not provided by the other three solvers, since they do not compute a roll-gap shape beyond the single working radius they read from `Roll.working_radius`.

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